CHEM 1301 Exam 3 — Master Study Guide

Chapter 7 (Reactions in Aqueous Solutions) · Chapter 8 (Chemical Composition) · Chapter 9 (Chemical Quantities)

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How to use this guide

- **Exam format:** 30–35 multiple-choice questions, choices a–e. You get a periodic table. **You do not get the solubility rules, so memorize Table 7.1** (Section 7.3).
- **Mix:** expect about half concept questions (definitions, classifications, "which is true") and half calculations.
- Each section lists the slides it comes from (`Chapter N post.pdf` / `.pptx`, same slide numbers). **Definitions are in the course's wording.** When a textbook or website words something differently, go with the slides.
- Every "Worked example" is a practice problem from the slides, solved step by step. Cover the solution, try it, then check.
- After each practice exam, use its key's "Source" line to jump back to the matching section here.

Chapter 7 — Reactions in Aqueous Solutions

7.1 Driving forces (Ch 7 slides 2–4)

A reaction tends to happen when it forms one of these. Memorize all four.

1. **Formation of a solid** (precipitation)
2. **Formation of water** (acid–base)
3. **Transfer of electrons** (oxidation–reduction)
4. **Formation of a gas** (gas evolution)

7.2 Ions in water and electrolytes (Ch 7 slides 6–10)

- When an ionic solid dissolves, **the ions separate and move around independently**. $Ba(NO_3)_2(aq)$ contains separated Ba^{2+} and NO_3^- ions (two NO_3^- for every Ba^{2+}), not $Ba(NO_3)_2$ units.
- Pure water does **not** conduct electricity. Ions must be present for current to flow.
- **Strong electrolyte:** a substance where every unit that dissolves produces separated ions.
- **A solid compound must have zero net charge**, so a precipitate must contain both a cation and an anion. Two cations (K^+ and Ba^{2+}) or two anions (CrO_4^{2-} and NO_3^-) can never form a solid together (slide 10).

7.3 Solubility rules — MEMORIZE (Ch 7 slide 11, Table 7.1)

#	Rule	Exceptions to know
1	Most **nitrate** (NO_3^-) salts are **soluble**	—
2	Most salts of **Na^+, K^+, NH_4^+** are **soluble**	—
3	Most **chloride** salts are **soluble**	**AgCl, $PbCl_2$, Hg_2Cl_2** are insoluble
4	Most **sulfate** salts are **soluble**	**$BaSO_4$, $PbSO_4$, $CaSO_4$** are insoluble
5	Most **hydroxides** are only **slightly soluble**	**NaOH, KOH** soluble; $Ba(OH)_2$, $Ca(OH)_2$ moderately soluble
6	Most **sulfide** (S^{2-}), **carbonate** (CO_3^{2-}), **phosphate** (PO_4^{3-}) salts are only **slightly soluble**	—

"Insoluble" and "slightly soluble" mean the same thing here: too little dissolves to see.

Rules 1 and 2 beat rules 5 and 6. If a compound contains Na^+, K^+, NH_4^+, or NO_3^-, it is soluble, even when the other ion is a sulfide, carbonate, phosphate, or hydroxide. Examples: $(NH_4)_2S$, K_3PO_4, and Na_2CO_3 are all **soluble**.

Memory aid, in three tiers:

- **Always soluble:** NO_3^-, Na^+, K^+, NH_4^+.

- **Soluble except:** Cl^- (except **Ag, Pb, Hg_2**), SO_4^{2-} (except **Ba, Pb, Ca**). Pb is on both exception lists.
- **Insoluble except tier 1:** OH^- (NaOH, KOH soluble), S^{2-}, CO_3^{2-}, PO_4^{3-}.

Worked example (slide 12). Classify NaOH, $BaSO_4$, $CaCO_3$.

- NaOH: **soluble**. It is a named exception in rule 5, and rule 2 covers Na^+.
- $BaSO_4$: **insoluble**. It is a named exception in rule 4.
- $CaCO_3$: **insoluble**. Rule 6 covers carbonates, and Ca^{2+} is not a rule-2 cation.

HW14 style: "Which rule explains why ammonium sulfide is soluble?" Answer: *most salts of NH_4^+ are soluble* (rule 2), **not** the sulfide rule.

7.4 Predicting precipitates (Ch 7 slide 14)

1. **Write the reactants as they actually exist** (separated ions).
2. **Consider the solids that could form** by **exchanging the anions**.
3. **Use the solubility rules** to decide whether either product is insoluble. If neither is, write **NR** (no reaction).

Worked example (slides 15–17). $K_2CrO_4(aq) + Ba(NO_3)_2(aq)$. The ions are K^+, CrO_4^{2-}, Ba^{2+}, NO_3^-. Swapping anions gives KNO_3 (soluble, rules 1 and 2) and $BaCrO_4$, which is the solid on the slides.

$$K_2CrO_4(aq) + Ba(NO_3)_2(aq) \rightarrow BaCrO_4(s) + 2KNO_3(aq)$$

NR example (HW15). $KNO_3(aq) + Na_2S(aq)$. Swapping gives K_2S and $NaNO_3$. Both contain rule-2 cations, so both are soluble and the answer is **NR**.

7.5 Three kinds of equations (Ch 7 slides 15–20)

Equation	Shows	Spectator ions?
Molecular	complete formulas of all reactants and products; doesn't show the actual forms in solution	n/a
Complete ionic	every **strong electrolyte** written as separated ions; **all** reactants and products included	included
Net ionic	**only the components that undergo a change** (are directly involved)	**left out**

- **Spectator ions:** ions that don't participate directly in the reaction. They appear unchanged on both sides.
- Write as ions: soluble ionic compounds (aq), strong acids, strong bases. Keep as full formulas: solids (s), water (l), gases (g), and weak acids such as HNO_2.

Worked example (slide 19). $Pb(NO_3)_2(aq) + 2KI(aq) \rightarrow PbI_2(s) + 2KNO_3(aq)$

- Complete ionic: $Pb^{2+}(aq) + 2NO_3^-(aq) + 2K^+(aq) + 2I^-(aq) \rightarrow PbI_2(s) + 2K^+(aq) + 2NO_3^-(aq)$

- Cancel the spectators K^+ and NO_3^-.
- Net ionic: $\mathbf{Pb^{2+}(aq) + 2I^-(aq) \rightarrow PbI_2(s)}$

Worked example (slide 20). Sodium chloride solution added to silver nitrate solution.

- Molecular: $NaCl(aq) + AgNO_3(aq) \rightarrow AgCl(s) + NaNO_3(aq)$. AgCl is a rule-3 exception.
- Complete ionic: $Na^+(aq) + Cl^-(aq) + Ag^+(aq) + NO_3^-(aq) \rightarrow AgCl(s) + Na^+(aq) + NO_3^-(aq)$
- Net ionic: $\mathbf{Ag^+(aq) + Cl^-(aq) \rightarrow AgCl(s)}$

Net ionic checklist: right states, the smallest whole-number coefficients, and both charge and atoms balanced.

7.6 Acids, bases, and acid–base reactions (Ch 7 slides 21–28)

Course definitions

- **Arrhenius acid:** a substance that produces $\mathbf{H^+}$ **ions** (protons) when dissolved in water.
- **Arrhenius base:** a substance that produces **hydroxide ions ($\mathbf{OH^-}$)** in water.
- **Strong acid:** a substance that **completely dissociates (ionizes)** in water. Each molecule breaks into H^+ plus an anion. Because strong acids are strong electrolytes that produce H^+, they are called strong acids.
- **Strong base:** a **metal hydroxide that is very soluble in water** and completely breaks up into separated ions. Strong bases are **strong electrolytes**.
- **Salt:** the ionic compound formed along with water in an acid–base reaction. It may stay dissolved or precipitate.

Memorize the lists:

- **Strong acids: HCl, $\mathbf{HNO_3}$, $\mathbf{H_2SO_4}$.** Anything else here (HNO_2, HF, $HC_2H_3O_2$, H_3PO_4) is not on the list.
- **Strong bases: NaOH, KOH.** (HW16 also uses CsOH, another very soluble group-1 hydroxide.)

The six principles (slides 24–26), condensed

1. The common strong acids are HCl, HNO_3, H_2SO_4.
2. A strong acid completely dissociates.
3. A strong base is a very soluble metal hydroxide (NaOH, KOH) that completely breaks into ions.
4. **Strong acid + strong base always has the same net ionic equation: $\mathbf{H^+(aq) + OH^-(aq) \rightarrow H_2O(l)}$.**
5. One product is always water. The other is a **salt** that stays dissolved and can be recovered by evaporating the water.
6. H^+ is the acidic ion and OH^- is the basic ion.

Naming the salt (HW16): the base's cation goes with the acid's anion. HI + NaOH gives **NaI**, NaOH + HNO_3 gives $\mathbf{NaNO_3}$, and HCl + CsOH gives **CsCl**.

Worked example (slide 27). Balance $H_3PO_4 + Ca(OH)_2 \rightarrow Ca_3(PO_4)_2 + H_2O$.

1. Balance Ca: 3 $Ca(OH)_2$.
2. Balance PO_4 as a unit: 2 H_3PO_4.
3. Balance H and O: 6 H^+ + 6 OH^- make 6 H_2O.

$$2H_3PO_4 + 3Ca(OH)_2 \rightarrow Ca_3(PO_4)_2(s) + 6H_2O(l)$$

$Ca_3(PO_4)_2$ is a phosphate (rule 6), so it precipitates. Phosphoric acid isn't on the strong-acid list, so leave it molecular in the ionic equations:

- Net ionic: $2H_3PO_4(aq) + 3Ca^{2+}(aq) + 6OH^-(aq) \rightarrow Ca_3(PO_4)_2(s) + 6H_2O(l)$

Worked example (slide 28). Nitrous acid + potassium hydroxide.

- Molecular: $HNO_2(aq) + KOH(aq) \rightarrow KNO_2(aq) + H_2O(l)$, which is already balanced.
- HNO_2 is not a strong acid, so it stays molecular. KOH and KNO_2 are written as ions.
- Complete ionic: $HNO_2(aq) + K^+(aq) + OH^-(aq) \rightarrow K^+(aq) + NO_2{}^-(aq) + H_2O(l)$
- Net ionic: $\mathbf{HNO_2(aq) + OH^-(aq) \rightarrow NO_2{}^-(aq) + H_2O(l)}$. Only a **strong** acid gives $H^+ + OH^- \rightarrow H_2O$.

7.7 Oxidation–reduction (Ch 7 slides 29–34, 38)

- **Oxidation–reduction (redox) reaction:** a reaction that involves a **transfer of electrons**. One species loses electrons and another gains them (HW16).
- **Principle 1: metal + nonmetal always makes an ionic compound and is always redox.** The metal loses electrons to become a cation, and the nonmetal gains them to become an anion.
- **Principle 2:** two nonmetals can also undergo redox. **For now, you recognize these by O_2 appearing as a reactant or product.** The compound formed by two nonmetals is not ionic.

How many electrons? Use the charges in the ionic product. The number of electrons lost per metal atom equals the cation's charge.

Reaction	Ions formed	Electrons
$2Li + F_2 \rightarrow 2LiF$ (slide 38)	Li^+, F^-	each Li loses 1; each F gains 1
$Ba + Cl_2 \rightarrow BaCl_2$ (HW16)	Ba^{2+}, Cl^-	Ba loses 2; Cl_2 gains 2 (1 per Cl)
$\mathbf{2Al + N_2 \rightarrow 2AlN}$ **(slide 33)**	Al^{3+}, N^{3-}	**each Al loses 3**; each N gains 3

Worked example (slide 34). Solid copper reacts with aqueous iron(II) chloride to form solid iron and aqueous copper(II) chloride.

$$Cu(s) + FeCl_2(aq) \rightarrow Fe(s) + CuCl_2(aq)$$

This is already balanced. Cu (0) becomes Cu^{2+}, so **Cu loses 2 e^-**. Fe^{2+} becomes Fe (0), so **Fe^{2+} gains 2 e^-**. Cl^- doesn't change; it is a spectator.

Class	How to spot it	Slide example

7.8 Classifying reactions (Ch 7 slides 35–45)

Class	How to spot it	Slide example
Precipitation	two solutions make a **solid**	$K_2CrO_4 + Ba(NO_3)_2 \rightarrow BaCrO_4(s) + 2KNO_3$
Double-displacement (double-exchange)	two anions **swap** partners	same as above
Acid–base	H^+ ends up in **water**	$HCl + KOH \rightarrow H_2O + KCl$
Oxidation–reduction	**electrons transferred** (metal + nonmetal, or O_2 involved)	$2Li + F_2 \rightarrow 2LiF$
Gas evolution	a **gas** forms (also acid–base here)	$2HCl + Na_2CO_3 \rightarrow CO_2(g) + H_2O + 2NaCl$
Single-replacement	**one** anion moves from one cation to another (always redox)	$Zn + 2HCl \rightarrow ZnCl_2 + H_2$
Combustion	reacts with **oxygen**, releasing energy so fast that a **flame** results	$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$
Synthesis (combination)	a compound forms from **simpler materials**	$2Na + Cl_2 \rightarrow 2NaCl$
Decomposition	a compound breaks into simpler compounds or elements, usually by **heat or electric current**	$2HgO \rightarrow 2Hg + O_2$

One reaction can belong to several classes (HW17):

- $C + O_2 \rightarrow CO_2$ is synthesis, combustion, and redox.
- $2Fe + 3Cl_2 \rightarrow 2FeCl_3$ is synthesis and redox, but **not** combustion (no O_2).
- Combustion and synthesis/decomposition with elements are all redox. Combustion also produces **heat/energy** (HW17).

Worked example (slide 44).

- $N_2 + 3H_2 \rightarrow 2NH_3$ is **synthesis**.
- $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$ is **combustion**.
- $2H_2O \rightarrow 2H_2 + O_2$ is **decomposition**.

Balancing from words (HW17):

- Phosphorus + chlorine: $P_4 + 6Cl_2 \rightarrow 4PCl_3$
- Ammonium nitrite decomposes: $NH_4NO_2 \rightarrow N_2 + 2H_2O$
- Hydrobromic acid decomposes: $2HBr \rightarrow H_2 + Br_2$

Chapter 8 — Chemical Composition

8.1 Counting by weighing; atomic mass units (Ch 8 slides 2–6)

- Objects that differ slightly can be **counted by weighing** if you know their **average mass** (the jellybean idea).
- **Atomic mass unit:** 1 amu = 1.66×10^{-24} g. Average atomic masses: H 1.008, C 12.01, N 14.01, O 16.00, Na 22.99, Al 26.98.
- **Key idea (slide 6):** samples whose masses are in the **same ratio as their atomic masses** contain the **same number of atoms**. So 26.98 g Al and 63.55 g Cu contain the same number of atoms.

8.2 The mole (Ch 8 slides 7–9)

- **Mole (course definition):** the number equal to the number of carbon atoms in **12.01 g of carbon**.
- **Avogadro's number:** 6.022×10^{23}. One mole of anything is 6.022×10^{23} units of it.
- 1 mol of any element has a mass in grams equal to its atomic mass (Table 8.2): Al 26.98 g, Au 196.97 g, Fe 55.85 g, S 32.07 g, B 10.81 g, Xe 131.3 g, and every one of these contains 6.022×10^{23} atoms.

8.3 The conversion map (Ch 8 slides 10–15)

$$\text{grams} \xrightarrow{\div \text{ molar mass}} \text{moles} \xrightarrow{\times\, 6.022\times10^{23}} \text{atoms or molecules}$$

Go backward by doing the opposite operations. **Moles are always the middle step.** For mg, μg, or kg, convert to grams first.

Worked example (slide 10). How many H atoms are in 0.500 g of H?

$$0.500 \text{ g H} \times \frac{1 \text{ mol H}}{1.008 \text{ g H}} = 0.496 \text{ mol H}; \qquad 0.496 \text{ mol} \times \frac{6.022 \times 10^{23} \text{ atoms}}{1 \text{ mol}} = 2.99 \times 10^{23} \text{ H atoms}$$

Worked examples (slides 11–15).

Slide	Problem	Setup	Answer
11	3.60 g Al → mol	3.60 ÷ 26.98	**0.133 mol**
12	544 g Ca → mol	544 ÷ 40.08	**13.6 mol**
13	0.133 mol Al → atoms	$0.133 \times 6.022 \times 10^{23}$	**8.01×10^{22} atoms**
14	13.6 mol Ca → atoms	$13.6 \times 6.022 \times 10^{23}$	**8.19×10^{24} atoms**
15	2.31×10^{22} atoms Au → g	$(2.31 \times 10^{22} \div 6.022 \times 10^{23}) \times 196.97$	0.0384 mol, so **7.56 g**

Reality check (slide 19): a sample smaller than 1 molar mass must be less than 1 mol and fewer than 6.022×10^{23} atoms. An atom count around 10^{-20} or 10^{50} means you flipped a step.

HW18 style: "A sample of Co contains 3.10×10^{24} atoms; a sample of Mn contains 4.10 mol. Which has more atoms?" Convert to the same unit: 4.10 mol $\times$ 6.022×10^{23} = 2.47×10^{24} atoms of Mn, so the **Co** sample has more atoms.

8.4 Conceptual problem solving (Ch 8 slides 16–19)

1. **Where are we going?** Read the problem, identify the final goal, and sort the facts given (draw it if it helps).
2. **How do we get there?** Work backward from the goal. In stoichiometry, always start with the balanced equation.
3. **Reality check:** does the answer make sense?

8.5 Molar mass (Ch 8 slides 21–29)

- **Molar mass:** the mass in grams of **1 mole** of a substance, found by **summing the masses of the component atoms**.
- **A subscript outside parentheses multiplies everything inside.** $Ba(OH)_2$ has 1 Ba, 2 O, and 2 H.

Worked examples

Slide	Compound	Sum	Molar mass
22	CH_4	12.01 + 4(1.008)	**16.04 g/mol**
23	NH_3	14.01 + 3(1.008)	**17.03 g/mol**
24	H_2SO_4	2(1.008) + 32.07 + 4(16.00)	**98.09 g/mol**
25	$C_6H_{12}O_6$	6(12.01) + 12(1.008) + 6(16.00)	**180.16 g/mol**
26	$Ba(OH)_2$	137.33 + 2(17.01)	**171.35 g/mol**

Mass and moles (slides 27–29)

Slide	Problem	Work	Answer
27	7.89 mol NaOH → g	7.89 × 40.00	**316 g**
28	982 g FeO → mol	982 ÷ 71.85	**13.7 mol**
29	9.43 mg $CaCl_2$ → mol	9.43×10^{-3} g ÷ 110.98	**8.50×10^{-5} mol**

HW19 style: write the formula first, then the molar mass. Iron(II) chloride = $FeCl_2$ (126.75 g/mol), copper(II) sulfate = $CuSO_4$ (159.62 g/mol), rubidium chloride = RbCl. **Molecules** = moles × 6.022 × 10^{23}. For "a sample of C_6H_6 containing 5.09 mol C", the number of molecules is 5.09 ÷ 6 = 0.848 mol C_6H_6, then × N_A.

8.6 Mass percent (Ch 8 slides 30–33)

$$\text{mass percent of element} = \frac{\text{mass of the element in 1 mol of compound}}{\text{mass of 1 mol of compound}} \times 100\%$$

Worked example (slide 31). Ethanol, C_2H_5OH, 46.07 g/mol:

- C: 24.02 / 46.07 × 100 = **52.14%**
- H: 6.048 / 46.07 × 100 = **13.13%**

- O: 16.00 / 46.07 × 100 = **34.73%**

These add to 100%.

Worked example (slide 32). H_3PO_4, 97.99 g/mol:

- H: 3(1.008) / 97.99 = **3.09%**
- P: 30.97 / 97.99 = **31.60%**
- O: 64.00 / 97.99 = **65.31%**

Worked example (slide 33). Magnesium iodite, $Mg(IO_2)_2$ (the iodite ion is IO_2^-), 342.11 g/mol:

- Mg: 24.31 / 342.11 = **7.11%**
- I: 253.80 / 342.11 = **74.19%**
- O: 64.00 / 342.11 = **18.71%**

Trap: count the atoms of that element in the formula. KNO_2 has 2 O, and PCl_5 has 5 Cl (HW20: P 14.87%, Cl 85.13%).

8.7 Empirical and molecular formulas (Ch 8 slides 34–50)

- **Empirical (simplest) formula:** the **smallest whole-number ratio** of atoms in the compound.
- **Molecular formula:** the **actual** formula, giving the composition of the molecules present.
- **Molecular formula = (empirical formula)$_\mathbf{n}$**, where n is a whole number. For example, glucose $C_6H_{12}O_6$ has the empirical formula CH_2O.
- **What you need to know (slide 47, HW21):** the empirical formula needs the composition by mass. The molecular formula also needs the **molar mass**.

Empirical formula practice (slide 40)

- $H_2O_2 \rightarrow$ **HO**
- $C_4H_8O_2 \rightarrow$ **C_2H_4O**
- $P_4O_{10} \rightarrow$ **P_2O_5**

Benzoic acid, $C_7H_6O_2$, is already in lowest terms (HW20).

Steps for the empirical formula (slides 42–43)

1. Get the **mass of each element** in grams. For percentages, **assume a 100 g sample**, so % becomes g.
2. Convert each mass to **moles**.
3. **Divide by the smallest number of moles.** If every result is a whole number, those are the subscripts.
4. If not, **multiply all of them by the smallest integer** that makes them whole. For example, 1.5 × 2 = 3 and 1.33 × 3 = 4.

Worked example (slides 35–37). 32.000% C, 6.714% H, 42.630% O, 18.660% N.

	g (per 100 g)	mol	÷ 1.333
C	32.000	2.664	2
H	6.714	6.648	5
O	42.630	2.664	2

N	18.660	1.333	1

The empirical formula is $\mathbf{C_2H_5O_2N}$.

Worked example (slide 38). C 9.94%, Cl 58.6%, F 31.4%. The moles are C 0.828, Cl 1.653, F 1.653. Dividing by 0.828 gives 1 : 2 : 2, so the formula is $\mathbf{CCl_2F_2}$.

Worked example (slides 45–46). 1.198 g C combines with 0.302 g H. C: 1.198 / 12.01 = 0.0998 mol. H: 0.302 / 1.008 = 0.300 mol. The ratio H/C = 3.00, so the formula is $\mathbf{CH_3}$.

HW21 style:

- Mg 20.19%, S 26.64%, O 53.17% gives $\mathbf{MgSO_4}$.
- Si 10.85%, Cl 27.40%, Br 61.75% gives $\mathbf{SiCl_2Br_2}$.

Combustion analysis (slide 44): burn a carbon-containing sample in excess O_2. The masses of the products give the percent C (and H), which leads to the empirical formula.

Molecular formula from molar mass (slides 47–49)

$$n = \frac{\text{molar mass}}{\text{empirical formula mass}}$$

Worked example (slide 49). Empirical formula $C_4H_5N_2O$, molar mass 194.14 g/mol. The empirical mass is 4(12.01) + 5(1.008) + 2(14.01) + 16.00 = 97.10. So n = 194.14 / 97.10 = 2, and the molecular formula is $\mathbf{C_8H_{10}N_4O_2}$.

HW21 style:

- CH with molar mass ≈ 26: CH has mass 13.02, so n = 2 and the formula is $\mathbf{C_2H_2}$.
- C_3H_7NO with molar mass 145–150: the empirical mass is 73.10, so n = 2 and the formula is $\mathbf{C_6H_{14}N_2O_2}$.

Why the molecular formula is more useful (slide 50): it tells you the actual number of atoms in a molecule. The empirical formula only gives the ratio, and many different compounds share the same empirical formula.

Chapter 9 — Chemical Quantities

9.1 What a balanced equation tells you (Ch 9 slides 2–5)

- A balanced equation gives the **identities** of the reactants and products and **how much** of each takes part.
- **Coefficients give relative numbers of molecules (or moles), never grams.** What matters is the *ratio* of the coefficients, not any single coefficient.
- $CO(g) + 2H_2(g) \rightarrow CH_3OH(l)$ can be read as 1 molecule + 2 molecules → 1 molecule, as 1 dozen + 2 dozen → 1 dozen, or as **1 mol + 2 mol → 1 mol** (Table 9.1).

Worked example (slide 4). $2Al_2O_3 \rightarrow 4Al + 3O_2$.

- From 4 molecules of Al_2O_3: $4 \times (3/2) =$ **6 molecules of O_2**.
- If 8 atoms of Al are produced: $8 \times (3/4) =$ **6 molecules of O_2**.

9.2 Mole ratios (Ch 9 slides 5–9)

Every pair of substances in a balanced equation gives a mole ratio. From $2H_2O \rightarrow 2H_2 + O_2$:

- 2 mol H_2O = 2 mol H_2
- 2 mol H_2O = 1 mol O_2
- 2 mol H_2 = 1 mol O_2

Worked example (slide 7). $N_2 + 3H_2 \rightarrow 2NH_3$. The ratios are 1 mol N_2 : 3 mol H_2, 1 mol N_2 : 2 mol NH_3, and 3 mol H_2 : 2 mol NH_3.

Worked example (slide 8). $N_2 + O_2 \rightarrow 2NO$. The ratios are 1 mol N_2 : 1 mol O_2, 1 mol N_2 : 2 mol NO, and 1 mol O_2 : 2 mol NO.

Worked example (slide 9). Ethane burns: $\mathbf{2C_2H_6 + 7O_2 \rightarrow 4CO_2 + 6H_2O}$. To balance, do C first, then H, then O. 7/2 O_2 isn't a whole number, so double everything.

Orienting the ratio (HW22): put the unit you **want** on top and the unit you **have** on the bottom. For $4B + 3O_2 \rightarrow 2B_2O_3$, going from mol B to mol O_2 uses (3 mol O_2 / 4 mol B).

9.3 Mole–mole conversions (Ch 9 slides 10–12)

$$\text{mol wanted} = \text{mol given} \times \frac{\text{coefficient wanted}}{\text{coefficient given}}$$

Worked examples

- **Slide 10.** $Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$. From 17.5 mol CO: $17.5 \times (2/3) =$ **11.7 mol Fe**.
- **Slide 11.** $2C_2H_6 + 7O_2 \rightarrow 4CO_2 + 6H_2O$. For 0.987 mol C_2H_6: $0.987 \times (7/2) =$ **3.45 mol O_2**.
- **Slide 12.** $NH_3 + NaOCl \rightarrow NaOH + NH_2Cl$, a 1 : 1 ratio. 1.32 mol NH_3 gives **1.32 mol NH_2Cl** ($\times$ 51.48 g/mol = 67.9 g).

9.4 Stoichiometry with masses (Ch 9 slides 13–20)

Stoichiometry: using a balanced chemical equation to calculate the relative masses of reactants and products. **The equation must be balanced first**, because the mole ratios come from the coefficients (slide 16).

Steps (slide 14)

1. Balance the equation.
2. Convert grams to moles.
3. Set up the mole ratio.
4. Calculate the moles of the substance you want.
5. Convert moles back to grams.

$$\text{g A} \xrightarrow{\div \text{MM}_A} \text{mol A} \xrightarrow{\times \frac{\text{coef B}}{\text{coef A}}} \text{mol B} \xrightarrow{\times \text{MM}_B} \text{g B}$$

Worked examples

Slide	Equation / question	Work	Answer
17	$2SO_2 + O_2 \rightarrow 2SO_3$; g SO_3 from 2.33 mol O_2	2.33 × 2 = 4.66 mol; × 80.07	**373 g**
18	$H_2SO_4 + 2KOH \rightarrow K_2SO_4 + 2H_2O$; mol H_2SO_4 for 19.9 g KOH	19.9 ÷ 56.11 = 0.3547 mol; × ½	**0.177 mol**
19	$4NH_3 + 5O_2 \rightarrow 4NO + 6H_2O$; g NH_3 for 54.7 g NO	54.7 ÷ 30.01 = 1.823 mol; × 4/4; × 17.03	**31.0 g**
20	$CuSO_4 + Fe \rightarrow FeSO_4 + Cu$; g $CuSO_4$ for 244 g Fe	244 ÷ 55.85 = 4.369 mol; × 1/1; × 159.62	**697 g**

HW23 style: $Fe + Cl_2 \rightarrow FeCl_3$ must be balanced first, giving $2Fe + 3Cl_2 \rightarrow 2FeCl_3$. 20.5 mg Fe is 0.367 mmol Fe, which makes 0.367 mmol $FeCl_3$ × 162.20 = **59.5 mg**. You can work in mg and mmol throughout.

9.5 Limiting reactant (Ch 9 slides 21–24)

- **Limiting reactant (limiting reagent):** the reactant that is **used up first**. It **controls how much product forms**.
- The other reactant is **in excess**: some of it is left over.

Method

1. Convert every reactant to **moles**.
2. Pick one reactant and use the mole ratio to find **how much of the other it needs**.
3. If you have **less** of the other than it needs, the other is limiting. Otherwise, the first one is.

(Alternatively, calculate the product each reactant could make. The **smaller** amount comes from the limiting reactant.)

Never compare grams, and never compare raw coefficients.

Worked example (slide 23). $3Mg + N_2 \rightarrow Mg_3N_2$, with 4.75 mol Mg and 2.11 mol N_2. The Mg needs 4.75 / 3 = 1.58 mol N_2, and there are 2.11 mol, so N_2 is in excess and **Mg is limiting**.

Worked example (slide 24). $4Al + 3O_2 \rightarrow 2Al_2O_3$, with 17.9 g Al and 64 g O_2. Al = 0.663 mol and O_2 = 2.0 mol. The Al needs $0.663 \times 3/4 = 0.498$ mol O_2, which is much less than 2.0, so **Al is limiting**. Note that Al has the *smaller* mass here, but that is not the reason it's limiting.

9.6 Theoretical yield (Ch 9 slides 25–27)

Theoretical yield: the amount of product **calculated from the limiting reactant**. It is the **maximum** amount of product that can be made from the amounts of reactants used.

Worked example (slide 26). $Pb(NO_3)_2 + 2NaCl \rightarrow PbCl_2 + 2NaNO_3$, with 237 g NaCl and excess $Pb(NO_3)_2$.

$$237 \text{ g} \div 58.44 = 4.055 \text{ mol NaCl}; \quad \times\tfrac{1}{2} = 2.028 \text{ mol } PbCl_2; \quad \times 278.1 = \mathbf{564\ g\ PbCl_2}$$

Worked example (slide 27). $2Na + Cl_2 \rightarrow 2NaCl$, with 68 g Na and 97 g Cl_2.

- Na = 2.958 mol and Cl_2 = 1.368 mol.
- The Cl_2 needs 2.736 mol Na, and you have 2.958, so **Cl_2 is limiting**.
- NaCl = $1.368 \times 2 \times 58.44 =$ **160 g**.

9.7 Actual and percent yield (Ch 9 slides 28–32)

- **Actual yield:** the amount of product **actually produced**. It is usually less than the theoretical yield, for example because of **side reactions** that consume reactants or products.
- **Percent yield** shows how efficient a reaction is:

$$\text{percent yield} = \frac{\text{actual yield}}{\text{theoretical yield}} \times 100\%$$

- **Can the actual yield be higher than the theoretical yield (slide 29)?** No. The theoretical yield is the maximum. A result over 100% means an error, such as a wet or impure product or a math mistake.

Worked example (slide 31). $H_2 + I_2 \rightarrow 2HI$, with 69.1 g H_2 and excess I_2; 134 g HI is actually produced.

$$69.1 \div 2.016 = 34.28 \text{ mol } H_2 \ \rightarrow\ 68.55 \text{ mol HI} \times 127.91 = 8768 \text{ g HI (theoretical)}; \qquad \frac{134}{8768} \times 100 = \mathbf{1.53}\%$$

Worked example (slide 32). $2KClO_3 \rightarrow 2KCl + 3O_2$, with 191 g $KClO_3$; 89.9 g KCl is actually produced.

$$191 \div 122.55 = 1.559 \text{ mol} \;\rightarrow\; 1.559 \text{ mol KCl} \times 74.55 = 116.2 \text{ g}; \qquad \frac{89.9}{116.2} \times 100 = \mathbf{77}.\mathbf{4}\%$$

HW24 style (two-part): find the theoretical yield, which may require finding the limiting reactant first ($Xe + 2F_2 \rightarrow XeF_4$), and then the percent yield. "Percentage of capacity reached" (the LiOH and CO_2 canister) uses the same formula: actual / theoretical $\times$ 100.

Common traps — check these before you bubble

1. **Solubility exceptions:** AgCl, $PbCl_2$, Hg_2Cl_2 and $BaSO_4$, $PbSO_4$, $CaSO_4$ are insoluble. $Na^+/K^+/NH_4{}^+/NO_3{}^-$ always win.
2. **Net ionic means no spectators.** A choice that still has Na^+ and $NO_3{}^-$ on both sides is the complete ionic equation.
3. **Precipitates are (s).** The same formula labeled (aq) is a wrong choice.
4. **Strong acids are HCl, HNO_3, H_2SO_4 only.** $HNO_2 \neq HNO_3$.
5. **Electrons per atom = ion charge**, not the total for the equation.
6. **More than one class can apply.** A single replacement is also redox. Burning in O_2 is combustion even when one product forms.
7. **mg $\rightarrow$ g first.** 1 mg = 10^{-3} g and 1 μg = 10^{-6} g.
8. **Subscripts outside parentheses multiply everything inside**: $Al_2(SO_4)_3$ has 12 O.
9. **Mass percent uses mass, not atom counts.** 1 P out of 8 atoms is not 12.5% P.
10. **Empirical formula:** divide by the *smallest* number of moles; multiply to clear 0.5, 0.33, or 0.25 fractions.
11. **Coefficients are moles, not grams.** Always convert to moles before using a ratio.
12. **Limiting reactant:** compare moles needed with moles available. Never compare grams or coefficients alone.
13. **Percent yield $\leq$ 100%.** Divide actual by **theoretical** yield, not by the mass of the reactant.

Glossary of course definitions

Term	Course definition (slide)
Precipitation / precipitate	Formation of a solid when two solutions are mixed; the solid is the precipitate (7: 5, 35)
Strong electrolyte	Each unit that dissolves produces separated ions (7: 7)
Molecular equation	Shows complete formulas of all reactants and products (7: 15)
Complete ionic equation	All strong electrolytes shown as ions; all reactants and products included (7: 16, 18)
Net ionic equation	Only the components directly involved / that undergo a change; no spectators (7: 17)
Spectator ions	Ions that don't participate directly in the reaction (7: 17)
Arrhenius acid / base	Produces H^+ / OH^- in water (7: 21, 22)
Strong acid	Completely dissociates (ionizes) in water: HCl, HNO_3, H_2SO_4 (7: 24)
Strong base	Metal hydroxide that is very soluble and completely breaks into ions: NaOH, KOH (7: 25)
Salt	Ionic compound formed with water in an acid–base reaction (7: 23, 26)
Oxidation–reduction	Reaction involving a transfer of electrons (7: 30)
Double-displacement	Two anions exchanged between cations (7: 36)
Single-replacement	One anion moves from one cation to another (7: 40)
Combustion	Reaction with oxygen that releases energy fast enough to make a flame (7: 41)

Term	Course definition (slide)
Synthesis (combination)	A compound forms from simpler materials (7: 42)
Decomposition	A compound breaks into simpler compounds or elements, by heat or current (7: 43)
Atomic mass unit	1 amu = 1.66×10^{-24} g (8: 5)
Mole	Number of C atoms in 12.01 g of C = 6.022×10^{23} (Avogadro's number) (8: 7)
Molar mass	Mass in grams of 1 mole of a substance; sum of the atomic masses (8: 22)
Mass percent	(Mass of element in 1 mol) / (mass of 1 mol) × 100% (8: 30)
Empirical formula	Smallest whole-number ratio of atoms (8: 39)
Molecular formula	Actual formula = $(\text{empirical})_{\text{n}}$ (8: 39, 48)
Stoichiometry	Using a balanced equation to calculate masses of reactants and products (9: 15)
Limiting reactant	The reactant used up first; it controls the amount of product (9: 21)
Theoretical yield	Product amount calculated from the limiting reactant; the maximum possible (9: 25, 28)
Actual yield	Product amount actually obtained (9: 28)
Percent yield	Actual / theoretical × 100% (9: 30)